BURGER'S EQ (SIMPLIFIED NAVIER-STOKES)

$$\rho\left(\underline{u}_t + \underline{u}\,\nabla\underline{u}\right) = -\nabla p + \underbrace{\mu\,\Delta\underline{u}} \quad (+f)$$

$\underline{u}$ = PARTICLE VELOCITY

p = PRESSURE

- If $\mu = 0$ & $p = \text{CONST}$

$$\Rightarrow \underline{u}_t + \underbrace{\underline{u}\,\nabla\underline{u}}_{\frac{1}{2}\nabla(\underline{u}^2)} = 0$$

(INVISCID BURGER'S EQ.)

$$F(\underline{u}) = \frac{1}{2}\underline{u}^2$$

- If $\mu \neq 0$ & $p = \text{CONST}$

$$\Rightarrow \underline{u}_t + \underline{u}\,\nabla\underline{u} = \frac{\mu}{\rho}\Delta\underline{u}$$

(VISCID BURGER'S EQ)

$$F(\underline{u}) = \frac{1}{2}\underline{u}^2 - \frac{\mu}{\rho}\nabla\underline{u}$$

- $u_t + u\,u_x = 0$

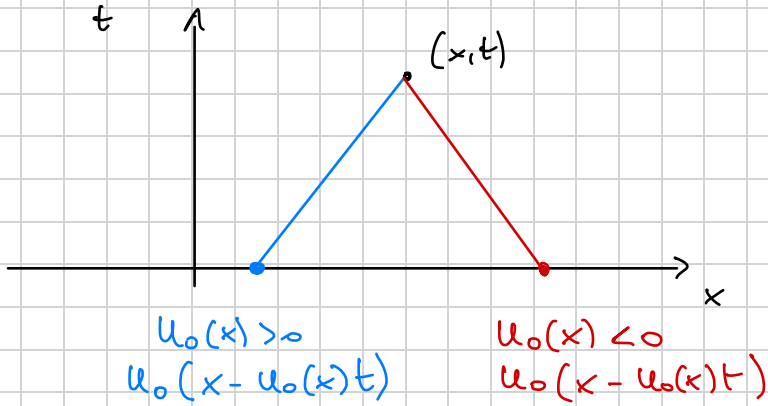
CHARACTERISTIC LINES

$$\begin{cases} x'(t) = u \\ x(0) = u_0(x) \end{cases}$$

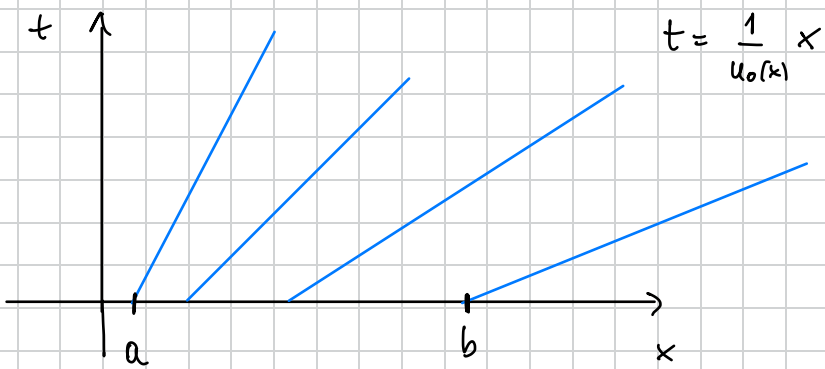
ALONG $x(t)$ u is constant

$u(x,t) = u_0(x-at)$ $\qquad$ $u(x,t) = u_0(x-ut)$

TRANSPORT EQ

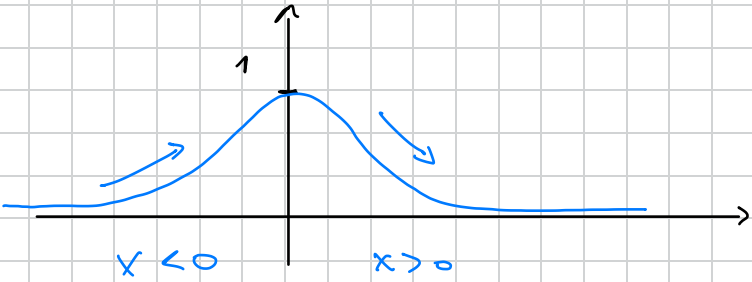


If $u_0(x) \nearrow$ $\Rightarrow$ $\exists!$ SOLUTION



WHAT HAPPENS IF $u_0(x) \nearrow\!\!\!\searrow$

Consider $u_0(x) = \dfrac{1}{1+x^2}$



Classical solutions exist only up to time

$$t \leq t_c = -\frac{1}{\inf_x u_0'(x)\, F''(u_0(x))}$$

$$F(u) = \frac{1}{2} u^2 \qquad u_0(x) = \frac{1}{1+x^2}$$

$$F'(u) = u \qquad F''(u) = 1$$

$$u_0'(x) = \frac{-1\,(2x)}{(1+x^2)^2} = -\frac{2x}{(1+x^2)^2}$$

$u_0''(x) = 0$ COMPUTE STATIONARY POINTS OF u_0'

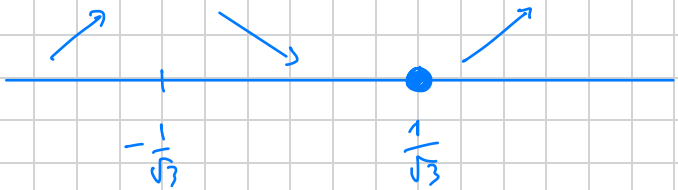
$$u_0''(x) = \frac{-2(1+x^2)^{\cancel{2}} + 2x\left(2\cancel{(1+x^2)}2x\right)}{(1+x^2)^{\cancel{4}\,3}}$$

$$= \frac{-2-2x^2+8x^2}{(1+x^2)^3} = \frac{6x^2-2}{(1+x^2)^3} = 0$$

$$3\cancel{6}x^2 = \cancel{2} \qquad x = \pm\sqrt{\frac{1}{3}}$$

$$u_0''(x) > 0 \quad \text{for} \quad x < -\sqrt{\frac{1}{3}} \ \vee \ x > \sqrt{\frac{1}{3}}$$

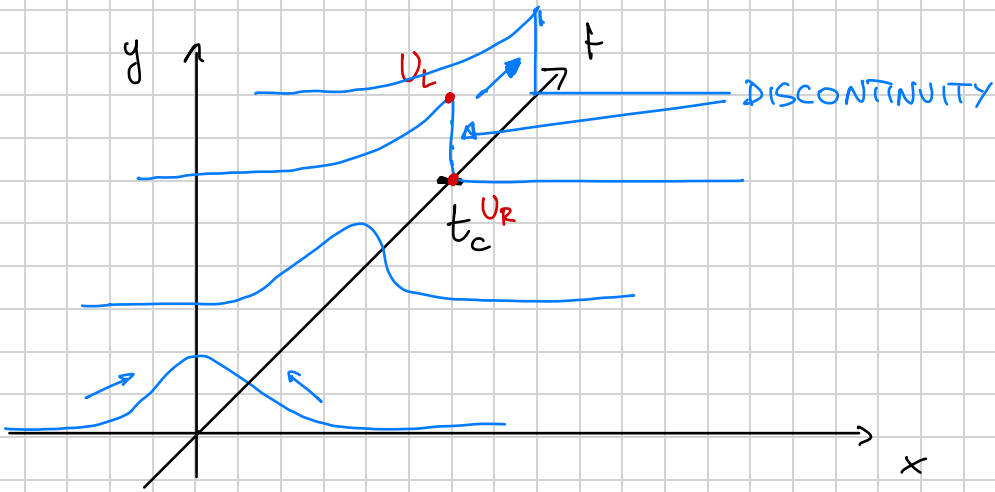
$u_0'(x)$ $\nearrow$



$$\inf_x u_0'(x) = u_0'\left(\frac{1}{\sqrt{3}}\right) = -\frac{2\cdot\frac{1}{\sqrt{3}}}{\left(1+\left(\frac{1}{\sqrt{3}}\right)^2\right)^2}$$

$$= \frac{-\frac{2}{\sqrt{3}}}{\left(\frac{4}{3}\right)^2} = -\frac{\cancel{2}}{\cancel{\sqrt{3}}}\,\frac{\cancel{9}\,3\sqrt{3}}{\cancel{16}\,8} = -\frac{3\sqrt{3}}{8}$$

$$t_c = -\frac{1}{-\frac{3\sqrt{3}}{8}\cdot 1} = \frac{8}{3\sqrt{3}}$$



WEAK SOLUTION : Solution to the Burger's eq (or other non-linear flux) s.t. satisfy this condition:

$$F(u_R) - F(u_L) = \sigma (u_R - u_L)$$

$$[F] = \sigma [u]$$

$$\sigma = \frac{[F]}{[u]} \rightarrow$$ SPEED OF PROPAGATION OF THE DISCONTINUITY

RANKINE - HUGONIOT CONDITION

WEAK SOLUTIONS ARE NOT UNIQUE.

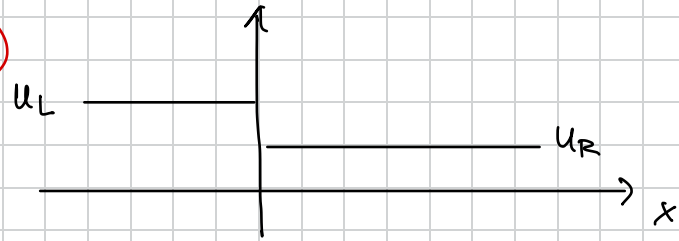
PHYSICAL SOLUTIONS ARE THE ENTROPIC ONES!

THAT SATISFY THE RH CONDITION.

EX: $F(u) = \frac{1}{2} u^2$ $\qquad u_0(x) = \begin{cases} u_L & x \leq 0 \\ u_R & x > 0 \end{cases}$

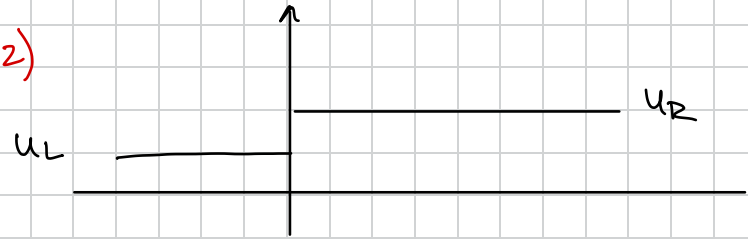
$u_L \neq u_R \qquad u_L, u_R > 0$

1)



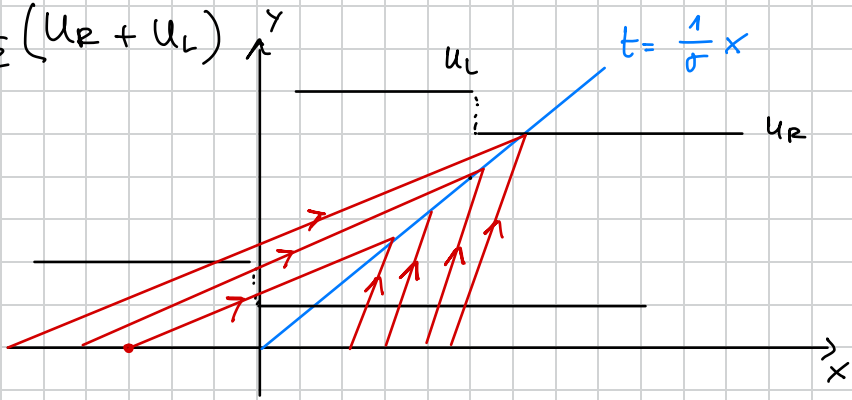
$u_L > u_R$

2)



1) $$\sigma = \frac{F(u_R) - F(u_L)}{u_R - u_L} = \frac{\frac{1}{2}\left(u_R^2 - u_L^2\right)}{u_R - u_L}$$

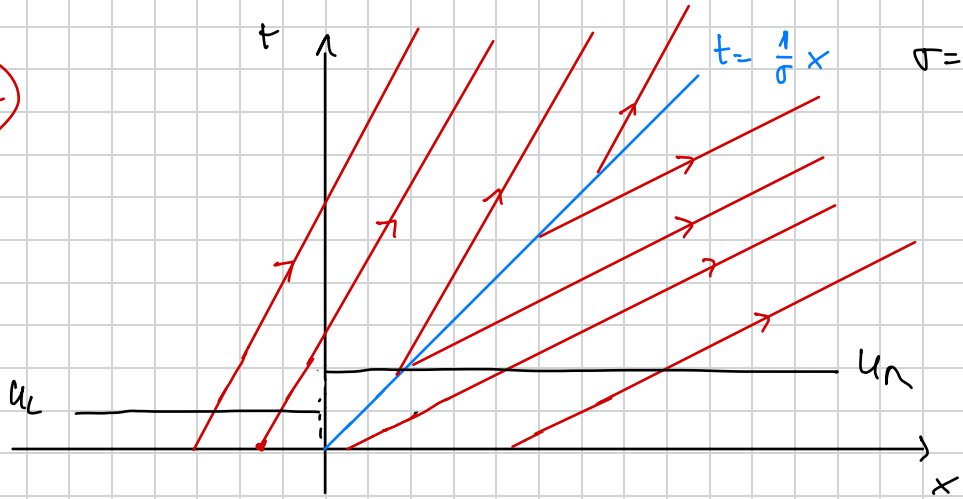
$$= \frac{1}{2}\left(u_R + u_L\right)$$



CHARACTERISTIC LINES ENTER THE DISCON.

$\exists!$ u SOLUTION $u(x,t) = \begin{cases} u_L & x \le \sigma t \\ u_R & x > \sigma t \end{cases}$

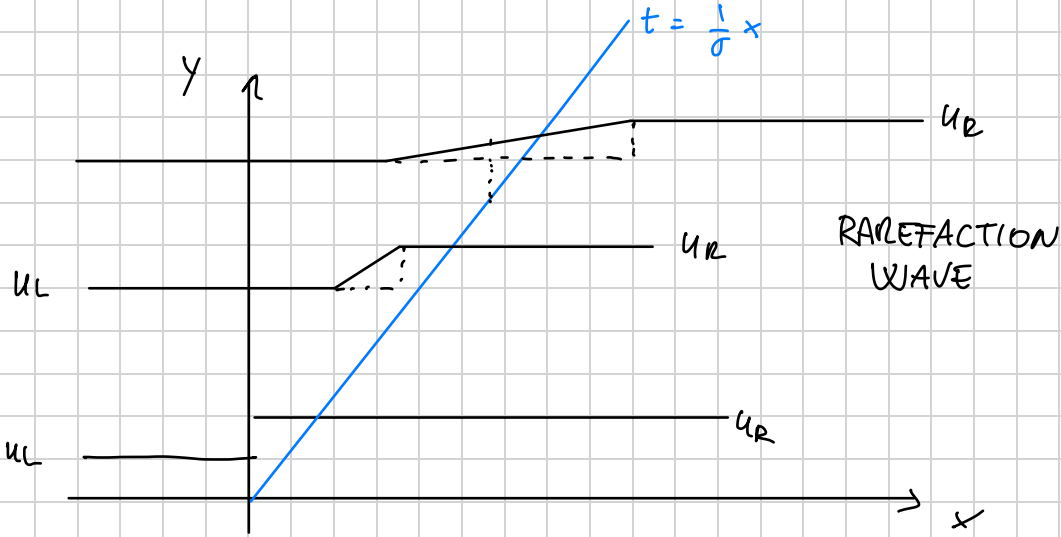
2)



$\sigma = \frac{1}{2}(u_L + u_R)$

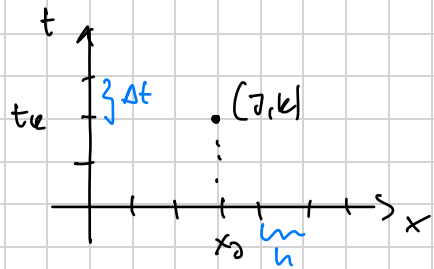
∞ SOLUTIONS! BUT ONLY ONE IS ENTROPIC STABLE

$$u(x,t) = \begin{cases} u_L & x < u_L t \\ x/t & u_L t \le x \le u_R t \\ u_R & x > u_R t \end{cases}$$



C) $$u_t + \left(\frac{1}{2}u^2\right)_x = 0$$

NC) $$u_t + u u_x = 0$$



UP WIND :

C) $$\begin{cases} U_j^{k+1} = U_j^k + \dfrac{\Delta t}{h}\left(\dfrac{U_j^{2k}}{2} - \dfrac{U_{j-1}^{2k}}{2}\right) & U_j^k > 0 \\ U_j^{k+1} = U_j^k + \dfrac{\Delta t}{h}\left(\dfrac{U_{j+1}^{k^2}}{2} - \dfrac{U_j^{k^2}}{2}\right) & U_j^k < 0 \end{cases}$$

NC) $$\begin{cases} U_j^{k+1} = U_j^k + \dfrac{\Delta t}{h} U_j^k \left(U_j^k - U_{j-1}^k\right) & U_j^k > 0 \\ U_j^{k+1} = U_j^k + \dfrac{\Delta t}{h} U_j^k \left(U_{j+1}^k - U_j^k\right) & U_j^k < 0 \end{cases}$$

LAX WENDROFF -(NC)

$$u_t + u u_x = 0$$

$$u_t + u u_x = \frac{\Delta t}{2} u_{tt} + \dots$$ 2ND ORDER TAYLOR EXP

$$u_{tt} + (u u_x)_t = 0$$

$$u_{tt} + u_t u_x + u u_{xt} = 0$$

$\rightarrow u_t = -u u_x$ $\qquad u_{xt} = (u_t)_x$

$$u_{tt} - u(u_x)^2 + u u_{xt} = 0 \qquad u_{tx} = -(u u_x)_x$$

$$u_{tt} = u(u_x)^2 + u(u u_x)_x$$

$$= u(u_x)^2 + u^2 u_{xx} + u(u_x)^2$$

$$= 2u(u_x)^2 + u^2 u_{xx}$$

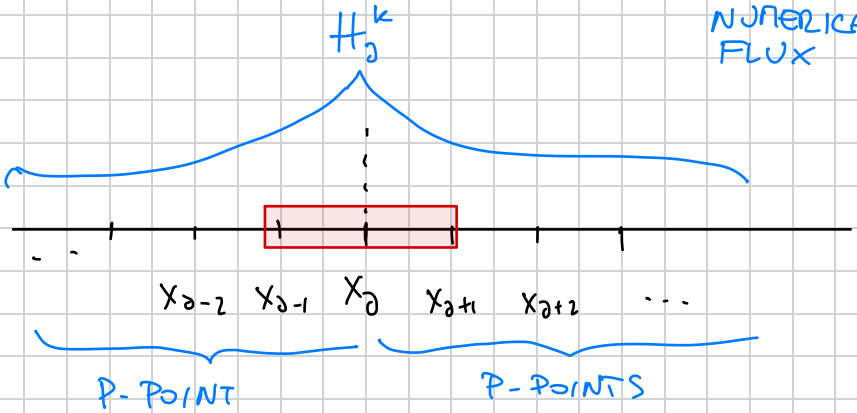
$$u_t + u u_x = \frac{\Delta t}{2}\left(2u(u_x)^2 + u^2 u_{xx}\right)$$

$$\overbrace{U_J^{k+1} = U_J^k}^{\Delta t\, u_t} - \Delta t\, \overbrace{U_J^k}^{u} \overbrace{\left(\frac{U_{J+1}^k - U_J^k}{2h}\right)}^{u_x}$$

$$+ \frac{\Delta t^2}{2}\left(\underbrace{2U_J^k}_{2u}\underbrace{\left(\frac{U_{J+1}^k - U_J^k}{h}\right)^2}_{u_x^2} + \underbrace{(U_J^k)^2}_{u^2}\underbrace{\frac{U_{J+1}^k - 2U_J^k + U_{J-1}^k}{h^2}}_{u_{xx}}\right)$$

CONSERVATIVE SCHEME FOR $U_t + F(u)_x = 0$

Based on $$U_j^{k+1} = U_j^k - \frac{\Delta t}{h}\left(\underbrace{H_{j-\frac{1}{2}}^k}_{\text{NUMERICAL FLUX}} - H_{j+\frac{1}{2}}^k\right)$$



$$H_{j+\frac{1}{2}}^k = H\left(U_{j-p}^k, \dots, U_{j-1}^k, U_j^k, U_{j+1}^k, \dots, U_{j+p}^k\right)$$

$$p=1 \qquad H_{j+\frac{1}{2}}^k = \underbrace{H\left(U_{j-1}^k, U_j^k, U_{j+1}^k\right)}_{?}$$

$$H_{j+\frac{1}{2}}^k = \underbrace{\frac{1}{\Delta t}\int_{t_k}^{t_{k+1}} F\left(U\left(x_{j+\frac{1}{2}}\right)\right) dt}_{\text{AT THE CONTINUOS LEVEL}}$$

(SAME IDEA OF LESO3)

LAX - FRIEDRICHS (c)

- $$U_j^{k+1} = \left(\frac{U_{j+1}^k + U_j^k}{2}\right) - \frac{\Delta t}{2h}\left(F(U_{j+1}^k) - F(U_{j-1}^k)\right)$$

$$F(u) = au$$

$$U_j^{k+1} = \frac{1}{2}\left(\frac{U_{j+1}^k + U_j^k}{2}\right) - \frac{\lambda}{2}\left(a u_{j+1}^k - a u_{j-1}^k\right)$$

LAX - WENDROFF (c)

SIMILAR TO

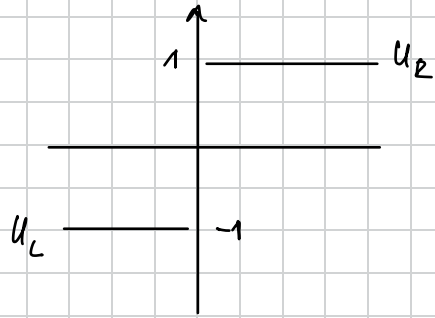
- $$U_{j+1}^k = U_j^k - \frac{\Delta t}{h}\left[\underbrace{F(U_{j+1}^k, U_j^k)}_{H_{j+\frac{1}{2}}^k} - \underbrace{F(U_j^k, U_{j-1}^k)}_{H_{j-\frac{1}{2}}^k}\right]$$

$$H(U_{j-1}^k, U_j^k, U_{j+1}^k)$$

$$F(U_{j+1}, U_j) = \frac{h}{2\Delta t}\left(U_j - U_{j+1}\right) + \frac{1}{2}\left(F(U_j) + F(U_{j+1})\right)$$

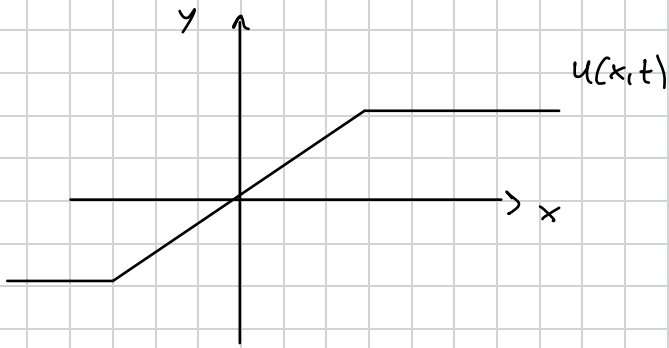
$$u_t + \left(\frac{u^2}{2}\right)_x = 0$$

$$u_0(x) = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases}$$



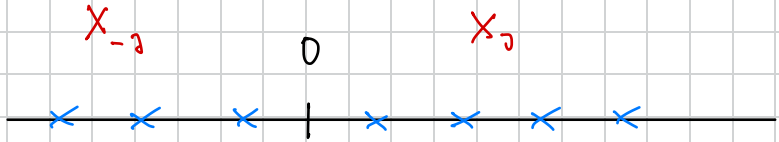
EXACT SOLUTION

$$u(x,t) = \begin{cases} 1 & x > t \\ \frac{x}{t} & -t \le x \le t \\ -1 & x < -t \end{cases}$$



BAD SOLUTION $\quad u(x,t) = u_0(x)$!

$$\sigma = \frac{[F]}{[u]} = \frac{1}{2}(u_L + u_R) = \frac{1}{2}(-1+1) = 0$$



$$\{x_j\}_{j=1}^{N} \not\ni \{x_0 = 0\}$$

$$u_j^0 = \begin{cases} -1 & x_j < 0 \quad j < 0 \quad (*) \\ 1 & x_j > 0 \quad j > 0 \quad (*) \end{cases}$$

$$U_j^{k+1} = U_j^k - \frac{\Delta t}{h}\left[F(U_{j+1}, U_j) - F(U_j, U_{j-1})\right]$$

$$F(v, w) = \begin{cases} F(v) & \sigma \geq 0 \\ F(w) & \sigma < 0 \end{cases}$$

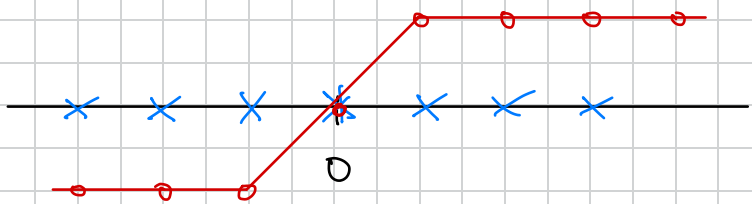
$$F(v, w) = F(v) = \frac{v^2}{2}$$

$$F(U_{j+1}, U_j) = \begin{cases} F(U_{j+1}, U_j) = F(-1, -1) = F(-1) = \frac{1}{2} \\ F(U_{j+1}, U_j) = F(1, -1) = \frac{1}{2} \\ F(U_{j+1}, U_j) = F(1, 1) = F(1) = \frac{1}{2} \end{cases}$$

$$U_j^{k+1} = U_j^k - \frac{\Delta t}{h}\left[\frac{1}{2} - \frac{1}{2}\right] = U_j^k$$

$$U_j^{k+1} = U_j^k = U_j^{k-1} = \dots = U_j^0$$

BAD SOLUTION



$$U_j^0 = \begin{cases} -1 & x_j < 0 \\ 0 & x_j = 0 \\ 1 & x_j > 0 \end{cases}$$

IDEA: Look for numerical scheme satisfy at the discrete level the entropic condition

↳ FINITE VOLUME SCHEMES !